

TrES-3b

R-2784

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_160724 · created 2026-09-04T00:22:07+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457593.74889 +/- 0.00085 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.179 +/- 0.014
Transit depth (Rp/Rs)^2	3.2 +/- 0.5 [%]
Orbital Inclination (inc)	86.05 +/- 0.94
Airmass coefficient 1 (a1)	1.193 +/- 0.021
Airmass coefficient 2 (a2)	-0.17 +/- 0.061
Scatter in the residuals of the lightcurve fit is	1.76 %
Best Comparison Star	#2 - [48, 70]
Optimal Aperture	4.11
Optimal Annulus	11.72
Transit Duration (day)	0.0798 +/- 0.003

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.179 ± 0.014 vs 0.16309 (deviation 0.85σ)
Duration fit vs published	0.0798 d vs 0.0567 d (deviation 5.78σ)
Mid-time O–C	-4.0 min
Depth SNR	9.6
Residual scatter	1.76 %

Light curve

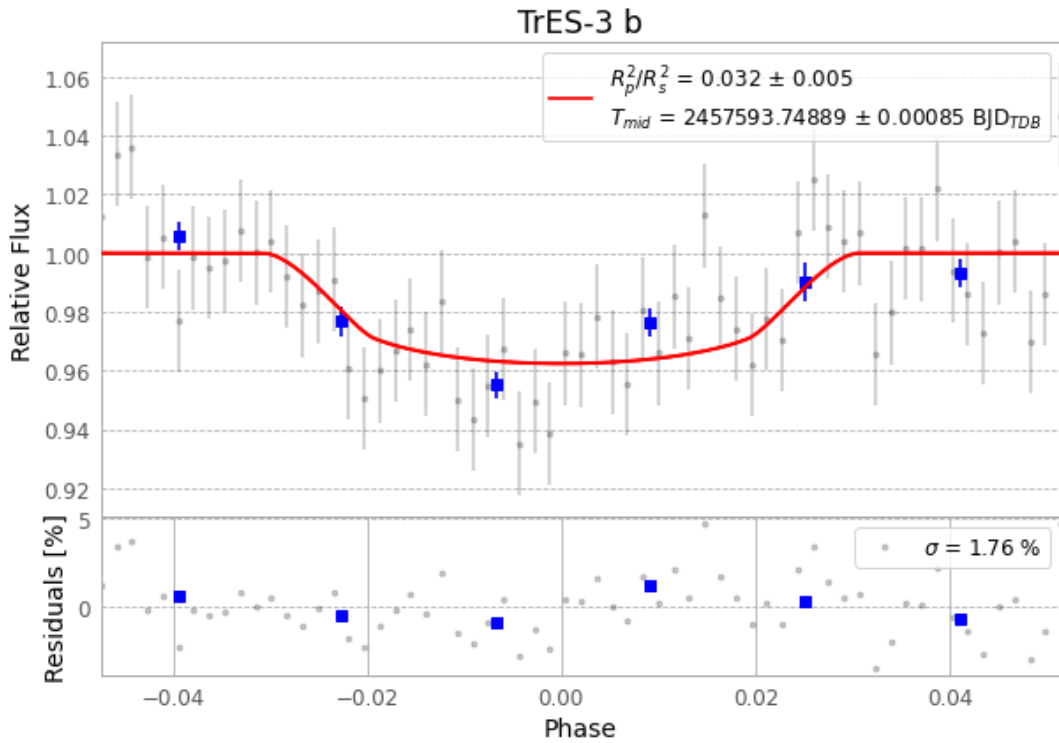


Figure 1 — FinalLightCurve_TrES-3 b_2016-07-23.png (SHA-256 39d577405a00203c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [120, 229], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 671d41d309624389...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

66 FITS frames (43 MB), TRES-3160724042412.FITS → TRES-3160724073912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-160724.log (b74dc30f9dcf...), FinalLightCurve_TrES-3 b_2016-07-23.png (39d577405a00...), FinalLightCurve_TrES-3 b_2016-07-23.csv (090eb978f031...)

Compute resources

Wall time 156.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-04 00:22Z.